

# Why have u written "p\_lower"? It appears nowhere in the section "2.6 Absolute Discounting"!!

You are completely right, and great catch! The notation  $p_{\text{lower}}$  does not literally appear anywhere in Section 2.6 of the paper [71–72].

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## What the Paper Actually Writes in Section 2.6

In Section 2.6 (Equation 18), Chen & Goodman express Absolute Discounting for an  $n$ -th order model as [71–72]:

$$p_{\text{abs}}(w_i \mid w_{i-n+1}^{i-1}) = \frac{\max\{c(w_{i-n+1}^i) - D, 0\}}{\sum_{w_i} c(w_{i-n+1}^i)} + (1 - \gamma_{w_{i-n+1}^{i-1}}) p_{\text{abs}}(w_i \mid w_{i-n+2}^{i-1})$$

where the scaling/interpolation term  $(1 - \gamma)$  is defined in Equation 19 as:

$$1 - \gamma_{w_{i-n+1}^{i-1}} = \frac{D}{\sum_{w_i} c(w_{i-n+1}^i)} N_{1+}(w_{i-n+1}^{i-1})$$

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## Why I Used $p_{\text{lower}}$ in the Handout

I wrote  $p_{\text{lower}}(w_i)$  as a **conceptual placeholder** for that last term:  $p_{\text{abs}}(w_i \mid w_{i-n+2}^{i-1})$ .

Here is how the paper's exact terms map to the concept:

1. **For a General  $n$ -gram Model:** The lower-order term in the paper is  $p_{\text{abs}}(w_i \mid w_{i-n+2}^{i-1})$ , which drops the oldest word  $w_{i-n+1}$  from the context history.
2. **For a Bigram Model ( $n = 2$ ):** The bigram history is  $w_{i-1}$ . Dropping that history leaves the unigram model, which the paper writes as  $p_{\text{abs}}(w_i)$  or  $p_{\text{ML}}(w_i)$ .

So  $p_{\text{lower}}(w_i)$  was just a descriptive label for **"whatever lower-order distribution we are dropping down to"** (the unigram  $p(w_i)$  in a bigram model).